
BINARY QUADRATIC FORMS

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§1 Introduction

1. A *binary quadratic form*¹ (or simply a *form*) is a polynomial of the form $q(X, Y) = aX^2 + bXY + cY^2$ with $a, b, c \in \mathbb{Z}$. We may think of it as a function $q : \mathbb{Z}^2 \rightarrow \mathbb{Z}$ that maps $(x, y) \in \mathbb{Z}^2$ to $ax^2 + bxy + cy^2 \in \mathbb{Z}$. We sometimes write $q = (a, b, c)$ for short.

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¹We shall sometimes write bqf for short

2 Definition

We make a few definitions. Let $q = (a, b, c)$ be a binary quadratic form.

- i. We say that q is *primitive* if $\gcd(a, b, c) = 1$.
- ii. We say that q *represents* the integer $n \in \mathbb{Z}$ if $q(x, y) = n$ for some $x, y \in \mathbb{Z}, (x, y) \neq (0, 0)$.
- iii. We say that q represents n *properly* or *primitively* if $q(x, y) = n$ for some $x, y \in \mathbb{Z}$ with $\gcd(x, y) = 1$. Otherwise the representation is *improper* or *imprimitively*. Note that $3^2 + 4^2 = 0^2 + 5^2 = 5^2$, so a given form may represent the same integer both primitively and imprimitively.
- iv. We define the *discriminant* of q to be $\text{disc}(q) = b^2 - 4ac$. An example is $\text{disc}(X^2 + 5Y^2) = \text{disc}(2X^2 + 2XY + 3Y^2) = -20$.

3 Proposition

Let $q = (a, b, c)$ be a binary quadratic form with $\text{disc}(q) = b^2 - 4ac = D$. Then

- i. q represents 0 if and only if D is a square,
- ii. $D \leq 0$ if and only if all non-zero values represented by q are of the same sign.

Proof. **Case I:** $a = 0$. In this case, $q(X, Y) = bXY + cY^2$ and $D = b^2$. Also, $q(x, 0) = 0$ for all integers x . This proves **i.** for this case. For **ii.**, observe that $D \leq 0$ is equivalent to $b^2 \leq 0$ which is obviously the same as saying $b = 0$. Thus, $q(X, Y) = cY^2$, and all its values are of the same sign.

Case II: $a \neq 0$. In this case, $4aq(X, Y) = (2aX + bY)^2 - DY^2$. Clearly, 0 is represented here if and only if $(2ax + by)^2 = Dy^2$ for some $(x, y) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$, which can hold if and only if D is a square. To see the truth of **ii.**, if $D \leq 0$, then clearly $4aq(x, y) \geq 0$ for all x, y . So, $q(x, y)$ is of constant sign (the same sign as a). For the reverse implication in **ii.**, suppose $q(x, y)$ is of constant sign. Then $4aq(-b, 2a) = -4a^2D$ and $4aq(1, 0) = 4a^2$. Since these two numbers are of the same sign, it follows that $-D \geq 0$, whence $D \leq 0$. $\square$

4 Proposition

Let $q = (a, b, c)$ be a binary quadratic form with $\text{disc}(q) = D < 0$. Then, for a given $n \in \mathbb{Z}$, there are at most a finite number of $x, y \in \mathbb{Z}$ such that $q(x, y) = n$.

This proposition is false for $D > 0$. For example, $X^2 - 2Y^2 = 1$ has infinitely many solutions. Indeed, for $n \geq 1$, write $(1 + \sqrt{2})^n = a_n + b_n\sqrt{2}$, where $a_n, b_n \in \mathbb{Z}$. Then $(1 - \sqrt{2})^n = a_n - b_n\sqrt{2}$ and so, $a_n^2 - 2b_n^2 = (-1)^n$ for all $n \geq 1$.

We now give a proof of the proposition.

Proof. Say $q(x, y) = n$. Then $4an = (2ax + by)^2 - Dy^2 \geq -Dy^2$, which implies $|y| \leq \sqrt{4an/-D}$, showing that the number of y 's is finite. Similarly, $(2ax + by)^2 \leq 4an \implies |2ax + by| \leq \sqrt{4an}$. Hence for each choice of y there are only a finite number of choices for x . $\square$

§2 Action of $\text{GL}_2(\mathbb{Z})$ on $Q(\mathbb{Z}^2)$

5 Notation. We denote the set of all binary quadratic forms by $Q(\mathbb{Z}^2)$. We also introduce the following notations, all of which are more or less standard:

$$M_2(\mathbb{Z}) := \{2 \times 2 \text{ matrices with entries from } \mathbb{Z}\};$$

$$\text{GL}_2(\mathbb{Z}) := \{A \in M_2(\mathbb{Z}) \text{ such that } A \text{ is invertible}\};$$

$$\text{SL}_2(\mathbb{Z}) := \{A \in \text{GL}_2(\mathbb{Z}) : \det A = 1\}.$$

We recall that $A \in M_2(\mathbb{Z})$ is said to be *invertible* if there is another matrix $B \in M_2(\mathbb{Z})$ such that $AB = I = BA$. Here, $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is the identity matrix. We remark that a 2×2 integer matrix is invertible if and only if its determinant is ± 1 . For A to be invertible, it is certainly *not* sufficient to have $\det A \neq 0$.

6. Let $q(X, Y) = aX^2 + bXY + cY^2 \in Q(\mathbb{Z}^2)$ and $A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in M_2(\mathbb{Z})$. We define the new binary quadratic form $q \cdot A$ by $(q \cdot A)(X, Y) = q(A \begin{pmatrix} X \\ Y \end{pmatrix}) = q(\alpha X + \beta Y, \gamma X + \delta Y)$. This is a linear change of variables. Note that

$$(q \cdot A)(X, Y) = a(\alpha X + \beta Y)^2 + b(\alpha X + \beta Y)(\gamma X + \delta Y) + c(\gamma X + \delta Y)^2.$$

which, when expanded, takes the form

$$= q(\alpha, \gamma)X^2 + (\dots)XY + q(\beta, \delta)Y^2 \quad (14:1)$$

where we have not shown the middle coefficient – later, we shall only need the coefficients of X^2 and Y^2 , which are respectively $q(\alpha, \gamma)$ and $q(\beta, \delta)$.

The following proposition is easy to prove:

7 Proposition

Let $q \in Q(\mathbb{Z}^2)$. Then

- i. $q \cdot I = q$,
- ii. $(q \cdot A) \cdot B = q \cdot (AB)$.

Proof. For **i.**, we have

$$(qI)(X, Y) = q(I\begin{pmatrix} X \\ Y \end{pmatrix}) = q\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} X \\ Y \end{pmatrix}\right) = q\left(\begin{pmatrix} X \\ Y \end{pmatrix}\right) = q(X, Y).$$

Part **ii.** can also be proved by elementary manipulations. $\square$

8. Let $G = \text{GL}_2(\mathbb{Z})$ or $\text{SL}_2(\mathbb{Z})$. Then G is clearly a group under the usual multiplication of matrices. The above proposition proves that there is a map

$$\begin{aligned} Q(\mathbb{Z}^2) \times G &\rightarrow Q(\mathbb{Z}^2), \\ (q, A) &\mapsto q \cdot A \end{aligned}$$

satisfying

- i. $q \cdot I = q$,
- ii. $(q \cdot A) \cdot B = q \cdot (AB)$.

for all $A, B \in G$ and all $q \in Q(\mathbb{Z}^2)$. Such a map is known as a (right) *action* of G on $Q(\mathbb{Z}^2)$. In general, given a group G and a nonempty set X , an *action* of G on X is a map

$$\begin{aligned} X \times G &\rightarrow X, \\ (x, g) &\mapsto x \cdot g \end{aligned}$$

satisfying $x \cdot e = x$ and $(x \cdot g) \cdot h = x \cdot (gh)$ for all $g, h \in G$ and all $x \in X$. This is a *right* action. Left actions are defined analogously. We shall sometimes conveniently drop the dot $\cdot$ in the action and write xg, qA etc in place of $x \cdot g, q \cdot A$ etc.

9 Definition [Equivalence]

Let $q_1, q_2 \in Q(\mathbb{Z}^2)$. We say that q_1 is *equivalent* to q_2 if there exists $A \in \text{GL}_2(\mathbb{Z})$ such that $q_2 = q_1 A$. We say that q_1 is *properly equivalent* to q_2 if there exists $A \in \text{SL}_2(\mathbb{Z})$ such that $q_2 = q_1 A$. We then write $q_1 \sim q_2$ for equivalence and $q_1 \stackrel{+}{\sim} q_2$ for proper equivalence.

10 Exercise

Prove that $\sim$ and $\overset{+}{\sim}$ are equivalence relations on $Q(\mathbb{Z}^2)$.

11 Example. Let $q(X, Y) = aX^2 + bXY + cY^2$ and take $A_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $A_2 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, $A_3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Let $q_i = qA_i$. Thus $q_1(X, Y) = q(A\begin{pmatrix} X \\ Y \end{pmatrix}) = q(X + Y, Y) = aX^2 + (2a + b)XY + (a + b + c)Y^2$ and you may similarly expand q_2 and q_3 . So we have

$$q = (a, b, c) \overset{+}{\sim} (a, 2a + b, a + b + c) = q_1.$$

Similarly, the second one gives us

$$q = (a, b, c) \overset{+}{\sim} (a, b - 2a, a - b + c) = q_2,$$

and the third one gives us

$$q = (a, b, c) \overset{+}{\sim} (c, -b, a) = q_3.$$

12 Definition

For a binary quadratic form $q = (a, b, c)$, we define its *matrix* by $\text{mat}(q) = \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}$. Clearly, $\det(\text{mat}(q)) = -\text{disc}(q)$.

13 Example. Find $\text{mat}(q \cdot A)$.

Solution. We claim that $\text{mat}(q \cdot A) = A^T \text{mat}(q) A$. It is easily verified that

$$2q(X, Y) = \begin{pmatrix} X & Y \end{pmatrix} \text{mat}(q) \begin{pmatrix} X \\ Y \end{pmatrix} \quad (14:2)$$

Using (14:2) and writing $A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, we have

$$\begin{aligned} \begin{pmatrix} X & Y \end{pmatrix} \text{mat}(q \cdot A) \begin{pmatrix} X \\ Y \end{pmatrix} &= 2(q \cdot A)(X, Y) \\ &= 2q\left(A\begin{pmatrix} X \\ Y \end{pmatrix}\right) \\ &= 2q(X', Y') \end{aligned}$$

where $X' = \alpha X + \beta Y$, $Y' = \gamma X + \delta Y$; in other words, we have just written

$$\begin{pmatrix} X' \\ Y' \end{pmatrix} = A \begin{pmatrix} X \\ Y \end{pmatrix}.$$

Note that the row vector $(X' \ Y')$ is then equal to the transpose of $A\begin{pmatrix} X \\ Y \end{pmatrix}$, i.e.,

$$\begin{aligned} (X' \ Y') &= \left(A \begin{pmatrix} X \\ Y \end{pmatrix} \right)^T \\ &= (X \ Y) A^T. \end{aligned} \quad (14:3)$$

Using (14:2) again as well as (14:3), we have

$$\begin{aligned} (X \ Y) \operatorname{mat}(q \cdot A) \begin{pmatrix} X \\ Y \end{pmatrix} &= 2q(X', Y') \\ &= (X' \ Y') \operatorname{mat}(q) \begin{pmatrix} X' \\ Y' \end{pmatrix} \\ &= (X \ Y) A^T \operatorname{mat}(q) A \begin{pmatrix} X \\ Y \end{pmatrix}, \end{aligned}$$

concluding the proof.

14 Theorem

Let $q_1 \sim q_2$. We then have

- i. $\operatorname{disc}(q_1) = \operatorname{disc}(q_2)$;
- ii. q_1 and q_2 represent the same set of integers the same number of times;
- iii. q_1 and q_2 are either both primitive or both imprimitive.

Proof. Say $q_2 = q_1 A$ with $A \in \operatorname{GL}_2(\mathbb{Z})$.

1. $\operatorname{disc}(q_2) = -\det \operatorname{mat}(q_2) = -\det(A^T(\operatorname{mat}(q_1))A) = -\det A^T \cdot \det \operatorname{mat}(q_1) \cdot \det A = -\det A^2 \cdot \det \operatorname{mat}(q_1) = -\det \operatorname{mat}(q_1) = \operatorname{disc}(q_1)$.
2. Say $q_2(x, y) = n$. Then $q_1\left(A\begin{pmatrix} x \\ y \end{pmatrix}\right) = n$. If $q_1(u, v) = m$ then $q_2(A^{-1}\begin{pmatrix} u \\ v \end{pmatrix}) = m$. Also, let $q_2(x, y) = n$ and $q_2(x', y') = n$ with $(x, y) \neq (x', y')$. Then $q_1\left(A\begin{pmatrix} x \\ y \end{pmatrix}\right) = m$ and $q_1\left(A\begin{pmatrix} x' \\ y' \end{pmatrix}\right) = m$ with $A\begin{pmatrix} x \\ y \end{pmatrix} \neq A\begin{pmatrix} x' \\ y' \end{pmatrix}$.
3. $q_1 = (a, b, c)$, primitive. $q_2 = (a', b', c')$, $\gcd(a', b', c') = d > 1$, and $q_2 = q_1 A$. Then d divides $q_2(x, y)$ for all integers x, y . Since the set of values of q_1 and q_2 are the same, it follows that $d \mid q_1(1, 0) = a$, $d \mid q_1(0, 1) = c$, $d \mid q_1(1, 1) = a + b + c \implies d \mid b$. Hence $d \mid a, b, c$ which is absurd since $\gcd(a, b, c) = 1$.

□

15 Theorem

Let $q \in Q(\mathbb{Z}^2)$ and $n \in \mathbb{Z}$. Then q properly represents n if and only if $q \stackrel{+}{\sim} (n, b, c)$ for some $b, c \in \mathbb{Z}$. Further, if $n \neq 0$, then we may take b such that $-|n| < |b| \leq |n|$.

Proof. Suppose $q \stackrel{+}{\sim} (n, b, c) = q'$. Say $q' = qA$ with $A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$. Then $n = q'(1, 0) = qA\begin{pmatrix} 1 \\ 0 \end{pmatrix} = q\begin{pmatrix} \alpha \\ \gamma \end{pmatrix} = q(\alpha, \gamma)$ and $\gcd(\alpha, \gamma) = 1$ since $\alpha\delta - \beta\gamma = 1$. So q properly represents n .

For the converse, suppose q properly represents n , say, $q(\alpha, \gamma) = n$ with $\gcd(\alpha, \gamma) = 1$. We want to show that $q \stackrel{+}{\sim} (n, b, c)$ for some $b, c \in \mathbb{Z}$. Now $\gcd(\alpha, \gamma) = 1$ implies that we can find $\beta, \delta \in \mathbb{Z}$ such that $\alpha\delta - \beta\gamma = 1$ (Bézout). Let $A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, and let $q' = q \cdot A$. Then $q'(1, 0) = (q \cdot A)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = q(\alpha, \gamma) = n$ which implies (see eq. (14:1)) $q' = (n, b, c)$ for some $b, c \in \mathbb{Z}$, completing the proof of the converse statement. It remains to prove the final assertion.

Thus, let $n \neq 0$. We can write $b = 2nu + r$ with $-|n| < r \leq |n|$. Let $q'' = q' \cdot B$ where $B = \begin{pmatrix} 1 & -u \\ 0 & 1 \end{pmatrix}$. This leads to $q''(X, Y) = (q' \cdot B)(X, Y) = nX^2 + rXY + c'Y^2$, completing the proof. $\square$

16. Let $D = b^2 - 4ac$ be the discriminant of $q = (a, b, c)$. Then $D \equiv b^2 \pmod{4}$ so $D \equiv 0$ or $1 \pmod{4}$. Conversely, if $D \equiv 0, 1 \pmod{4}$ then D is the discriminant of $X^2 - (D/4)Y^2$ if $D \equiv 0 \pmod{4}$ and of $X^2 + XY + ((1-D)/4)Y^2$ if $D \equiv 1 \pmod{4}$. These are known as the *principal forms* of discriminant D .

We now come to one of the important theorems of this chapter, due to Lagrange.

17 Theorem [Lagrange]

Let $D, n \in \mathbb{Z}$ with D not a square and $n \neq 0$. The following are equivalent:

- i. D is a square modulo $4n$.
- ii. There is a binary quadratic form q of discriminant D which properly represents n .

Proof. **i. $\implies$ ii.** There is some $b \in \mathbb{Z}$ such that $b^2 \equiv D \pmod{4n}$, say $b^2 = D + 4nc$ for some $c \in \mathbb{Z}$. This gives $b^2 - 4nc = D$. Let $q = (n, b, c)$. Then $q(1, 0) = n$ so q properly represents n .

ii. $\implies$ i. Let q be a binary quadratic form of discriminant D which properly represents n . Then by Theorem 15, we have $q \stackrel{+}{\sim} (n, b, c)$ for some $b, c \in \mathbb{Z}$. Since equivalent forms have the same discriminant, we see that D is also the discriminant of (n, b, c) , i.e., $D = b^2 - 4nc$ which gives us that $b^2 \equiv D \pmod{4n}$. Hence, D is a square modulo $4n$. $\square$

18 Corollary

Let $D \equiv 0$ or $1 \pmod{4}$ and let p be a prime such that $\left(\frac{D}{p}\right) = 1$. Then there is a binary quadratic form q of discriminant D which properly represents p . This form is unique up to equivalence.

Proof. By Lagrange, there is a binary quadratic form q which properly represents p . Then we know that $q \stackrel{+}{\sim} (p, b, c)$ for some $b, c \in \mathbb{Z}$ with $-p < b \leq p$. We also have $(p, b, c) \sim (p, -b, c)$, via the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. So, we may choose $0 \leq b \leq p$. Thus, $q \sim (p, b, c)$ with $0 \leq b \leq p$.

To prove uniqueness, let q' be another form of discriminant D which properly represents p . As above, we may take $q' \sim (p, b', c')$ with $0 \leq b' \leq p$. We shall prove that $b = b'$ and $c = c'$.

Since (p, b, c) and (p, b', c') have the same discriminant, we have $b^2 - 4pc = b'^2 - 4p c'$ so $4p \mid (b - b')(b + b')$. In particular, both $b + b'$ and $b - b'$ are even, and p must divide at least one of $b + b'$ and $b - b'$. However, if $p \mid b - b'$ then we are done: $b = b'$ (as $b - b'$ is even, it cannot be equal to $\pm p$!) and hence $c = c'$. So assume $p \mid b + b'$. Since $0 \leq b, b' \leq p$, the only possibilities are $b + b' = 0$ and $b + b' = 2p$ (since both $b + b'$ and $b - b'$ are even!). In the former case, $b = b' = 0$ and in the latter, $b = b' = p$. In both cases, the conclusion is that $b = b'$ and hence $c = c'$. This completes the proof. $\square$

§3 Discriminants

19. We shall classify binary quadratic forms based on their discriminant. First, a definition. An integer $D \in \mathbb{Z}$ is called a *discriminant* if it is the discriminant of some binary quadratic form. Recall that D is a discriminant if and only if $D \equiv 0, 1 \pmod{4}$. We say that a discriminant D is a *fundamental* discriminant if every binary quadratic form of discriminant D is primitive.

For example, 5 is a fundamental discriminant. For if $q = (a, b, c)$ is of discriminant 5, and if $d = \gcd(a, b, c)$, then $d^2 \mid b^2 - 4ac = 5$, which implies $d = 1$, showing that (a, b, c) is primitive.

On the other hand, 4 is not a fundamental discriminant, but 8 is. For if (a, b, c) has discriminant 8 then $b^2 - 4ac = 8$. Let d be the gcd of (a, b, c) ; we then have $d^2 \mid 8$. Suppose $d = 2$. In that case, the form is $(2a', 2b', 2c') \implies 4b'^2 - 16a'c' = 8 \implies b'^2 - 4a'c' = 2$. This is not possible as 2 is not a discriminant.

20 Proposition

Let D be a discriminant. Then D is a fundamental discriminant if and only if

- i. *D is of the form $4k + 1$, and D is square free, or*
- ii. *D is of the form $4k$, and $D/4$ is square free, and $D/4$ is of the form $4k + 3$ or $4k + 2$.*

Proof. Let D be a fundamental discriminant. We consider the two cases separately.

Case I: D is of the form $4k + 1$. We claim that D is square free. Suppose $D = r^2s$ with s being square free. If $r \neq 1$, then $rX^2 + rXY + r\frac{1-s}{4}Y^2$ is an imprimitive binary quadratic form of discriminant, D which contradicts the fact that D is fundamental. Note that $(1-s)/4$ is an integer: Indeed, $r^2s = D \equiv 1 \pmod{4}$ and $r^2 \equiv 0, 1 \pmod{4}$. But $r^2 \equiv 0 \pmod{4}$ is not possible, so $r^2 \equiv 1 \pmod{4}$ and we have $s \equiv 1 \pmod{4}$.

Case II: D is of the form $4k$. In this case, we must prove that $D/4$ is squarefree and $D/4 \equiv 2$ or $3 \pmod{4}$.

Let us first prove that $D/4$ is squarefree. Write $D/4 = r^2s$ where r, s are integers with s squarefree. If $r \neq 1$, then the form $rX^2 - rsY^2$ has discriminant $4r^2s = D$ and is not primitive, contradicting the fundamentality of D .

Next we prove that $D/4 \equiv 2$ or $3 \pmod{4}$. If $D/4 \equiv 0 \pmod{4}$, then $2X^2 - (D/8)Y^2$ would be an imprimitive binary quadratic form with discriminant D , contrary to our assumption that D is fundamental. Hence $D/4$ is not divisible by 4. Thus the only possibilities are $D/4 \equiv 1, 2$ or $3 \pmod{4}$.

Now let us suppose $D/4 \equiv 1 \pmod{4}$. Then $2X^2 + 2XY + ((4-D)/8)Y^2$ is a binary quadratic form with discriminant D , and is not primitive, a contradiction again. So, $D/4$ must leave a residue of 2 or 3 modulo 4.

□

§4 Class number

21. Let us first answer the question: What are all the binary quadratic forms of discriminant 0? Can we list all of them?

Suppose $q \in Q(\mathbb{Z}^2)$ has discriminant 0. Say $q = (a, b, c)$. Let us break into two cases.

Case I: $a = 0$. In this case, $0 = D = b^2 - 4 \cdot 0 = b^2$ so $b = 0$. Hence, we have $q = (0, 0, c)$.

Case II: $a \neq 0$. Then $4aq(x, y) = (2ax + by)^2 - Dy^2 = (2ax + by)^2$, so $q(-b, 2a) = 0$ with $(-b, 2a) \neq (0, 0)$. Thus, q represents 0, so by Theorem 15, $q \sim^+ (0, b', c')$ for some $b', c' \in \mathbb{Z}$. Again $b'^2 = 0$ gives $b' = 0$, i.e., $q \sim^+ (0, 0, c')$.

Thus, in both cases, we have proved that $q \sim^+ (0, 0, c)$ for some $c \in \mathbb{Z}$. The question now is, for $c \neq c'$, are the two forms $(0, 0, c)$ and $(0, 0, c')$ equivalent, properly or improperly?

Suppose $c \neq c'$ and let $q = (0, 0, c)$ and $q' = (0, 0, c')$. Say $c < c'$ to fix ideas. If both c and c' are nonzero, then $c \notin \{c'y^2 \mid y \in \mathbb{Z}\}$. By Theorem 14, the two forms q and q' are not equivalent. If one of them is 0, say $c = 0$ but $c' \neq 0$, then $\{cy^2\} = \{0\} \neq \{c'y^2 \mid y \in \mathbb{Z}\}$. Again by Theorem 14, they are not equivalent.

The upshot of the above discussion is that there are infinitely many binary quadratic forms of discriminant 0 which are inequivalent, namely all the forms $(0, 0, c)$ as c runs through the integers.

22. Having settled the case of discriminant 0, we now look at forms of nonzero discriminant. Here, it is convenient to consider the case $D = d^2$ first.

23 Proposition

Let $D = d^2 \neq 0$ with $d > 0$. Then, up to proper equivalence, there are only finitely many distinct binary quadratic forms of discriminant D .

Proof. Let q be of discriminant $D = d^2 \neq 0$. Hence, q properly represents 0, and so, $q \sim^+ (0, b, c)$ for some $b, c \in \mathbb{Z}$. Thus $b^2 = D = d^2 \implies b = \pm d \neq 0$. Write $c = bu + r$ with $0 \leq r < |b|$. Then $(0, b, c) \sim^+ (0, b, c - bu) = (0, b, r)$ via the matrix $\begin{pmatrix} 1 & -u \\ 0 & 1 \end{pmatrix}$. Hence, there are at most $|b| = d$ distinct forms up to proper equivalence. $\square$

24. The above proposition completely classifies all binary quadratic forms whose discriminants are squares. We now look at the other case, namely, when D is not a square. The following lemma, known as Lagrange Reduction Lemma, will help us classify such forms.

25 Lemma [Lagrange Reduction Lemma]

Let q be a binary quadratic form of discriminant $0 \neq D \in \mathbb{Z}$ which is not a square. Then q is properly equivalent to a form (a, b, c) with $|b| \leq |a| \leq |c|$. Such a form (a, b, c) further satisfies $1 \leq |a| \leq \sqrt{|D|/3}$.

Proof. Let $V = \{|q(x, y)| : \gcd(x, y) = 1\}$. Since D is not a square, $0 \notin V$. Let $a = q(x_1, y_1)$ be such that $|a| = \min V$. Since q represents a , we have $q \stackrel{+}{\sim} (a, b, c)$ for some $b, c \in \mathbb{Z}$ with $-|a| < b \leq |a|$ (by Theorem 15). This implies $|b| \leq |a|$. Since q also represents c , we have $|a| \leq |c|$. Thus $|b| \leq |a| \leq |c|$.

Now $b^2 - 4ac = D$ implies $|D| \geq 4|ac| - |b|^2 \geq 4|a|^2 - |a|^2 = 3|a|^2$ so $1 \leq |a| \leq \sqrt{|D|/3}$, proving the final assertion of the lemma. $\square$

26 Theorem [Lagrange]

Let $D \in \mathbb{Z}$ be a discriminant $D \neq 0$ and D not a square. Then up to proper equivalence, there are only finitely many distinct binary quadratic forms of discriminant D .

Proof. We know that $q \stackrel{+}{\sim} (a, b, c)$ where $|b| \leq |a| \leq |c|$ and $1 \leq |a| \leq \sqrt{|D|/3}$. Hence there are only finitely many possibilities for a and hence for b , as $|b| \leq |a|$. Since $b^2 - 4ac = D$, there are also only finitely many options for c . $\square$

We can now make the following definition:

27 Definition

Let $D \neq 0$ be a discriminant. $\mathcal{C}\ell(D)$ is the set of proper equivalence classes of binary quadratic forms of discriminant D , and $\mathcal{P}(D)$ is set of proper equivalence classes of *primitive* binary quadratic forms of discriminant D . We write $h(D) = |\mathcal{P}(D)|$. This is known as the *class number* of D .

28 Example. As an example, let us find $h(-4)$. Suppose q is a primitive binary quadratic form of discriminant -4 . Then $q \stackrel{+}{\sim} (a, b, c)$ with $|b| \leq |a| \leq |c|$. Recall that we always assume $a > 0, c > 0$ when the discriminant is negative. So $|b| \leq a \leq c$. Since $b^2 \equiv D \pmod{4}$, we have $b \equiv 0 \pmod{2}$ and $|b| \leq a \leq \sqrt{4/3} < \sqrt{2}$. Thus the only possibility is $b = 0$. Hence

$(a, b, c) = (a, 0, c)$. Since $b^2 - 4ac = -4$ we also have $a = c = 1$, so $q \stackrel{+}{\sim} (1, 0, 1)$. In other words, any primitive binary quadratic form of discriminant -4 is equivalent to $X^2 + Y^2$, so $h(-4) = 1$.

Also recall that if $\left(\frac{D}{p}\right) = 1$, then q primitively represents p , i.e., if $\left(\frac{-4}{p}\right) = 1$ then there exist $x, y \in \mathbb{Z}$ such that $x^2 + y^2 = p$. Now $\left(\frac{-4}{p}\right) = \left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = 1$ if and only if $p \equiv 1 \pmod{4}$. Thus every prime $p \equiv 1 \pmod{4}$ can be written as a sum of squares.

29 Exercise

Prove that $h(D) = 1$ for $|D| \leq 11$ with D not a square. Thus the only D 's to consider are $D = -11, -8, -7, -4, -3, 5, 8$.

A more complete list of negative discriminants with class number 1 is provided by the following theorem of Gauss:

30 Theorem [Gauss]

Let

$$D \in \{-3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163\}.$$

Then $h(D) = 1$.

The above theorem was proved by Gauss. He also conjectured that these are the only values of $D < 0$ for which the class number is 1. This was finally proved by Heegner in 1952. In his honour, the numbers 1, 2, 3, 7, 11, 19, 43, 67, 163 are known as *Heegner numbers*. They are exactly the positive squarefree integers d such that the ring of integers of the quadratic number field² $\mathbb{Q}[\sqrt{-d}]$ has unique factorisation. His proof was suspected to be erroneous for a long time, until it was finally understood and clarified by Stark in 1967. An independent proof was given by Baker around 1966.

Note that of the discriminants in Gauss' Theorem 24, only $-3, -4, -7, -8, -11, -19, -43, -67, -163$ are fundamental discriminants, and the other four $-12, -16, -27, -28$ are non-fundamental.

Here is a well-known open conjecture regarding class numbers:

²In this course, we will not be able to clarify the link with imaginary quadratic fields.

31 Conjecture

Are there infinitely many $D > 0$ such that $h(D) = 1$?

§5 Gaussian reduction

32. From now on, we shall assume that our binary quadratic forms are of negative discriminant, with positive values, i.e., $q = (a, b, c)$ with $D = b^2 - 4ac < 0$, $a > 0, c > 0$.

33 Definition

A binary quadratic form $q = (a, b, c)$ with discriminant $D = b^2 - 4ac < 0$ is said to be *reduced* (in the sense of Gauss) if either

$$-a < b \leq a < c$$

or

$$0 \leq b \leq a = c.$$

34 Example. For example, the forms $(3, 1, 4)$ and $(2, 1, 2)$ are reduced, whereas the forms $(1, 2, 1)$ and $(2, -1, 2)$ are not reduced.

The question which will interest us is: Given a non-reduced form, can we always bring it to a reduced form? Before proving the general theorem in this direction, let us look at some examples.

35 Example. i. Let us reduce $(3, 2, 1)$. We have

$$\begin{aligned} (3, 2, 1) &\stackrel{+}{\sim} (1, -2, 3) \text{ via } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &\stackrel{+}{\sim} (1, 0, 2) \text{ reduced, via } \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

ii. Similarly, we reduce $(4, 5, 5)$ —

$$\begin{aligned} (4, 5, 5) &\stackrel{+}{\sim} (4, -3, 4) \text{ via } \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \\ &\stackrel{+}{\sim} (4, 3, 4) \text{ reduced, via } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \end{aligned}$$

36. An integer vector $u = (x, y) \in \mathbb{Z}^2$ will be called *primitive* if $\gcd(x, y) = 1$. The following lemma plays a crucial role in our proof of the uniqueness of reduced form.

37 Lemma

Let $q = (a, b, c)$ be a reduced binary quadratic form of discriminant $D = b^2 - 4ac < 0$, $a > 0$, $c > 0$. Then a is the smallest value represented by q .

- **Case I:** If $a < c$, there are exactly two primitive vectors $u \in \mathbb{Z}^2$ such that $a = q(u)$, namely $u = (\pm 1, 0)$, and exactly two primitive vectors $v \in \mathbb{Z}^2$ such that $c = q(v)$, namely $v = (0, \pm 1)$, unless $a = b$, in which case, there are four vectors $v \in \mathbb{Z}^2$ such that $c = q(v)$, namely $v = (0, \pm 1), (1, -1), (-1, 1)$.
- **Case II:** If $a = c$, then there are exactly four primitive vectors $u \in \mathbb{Z}^2$ such that $a = q(u)$, namely $u = (\pm 1, 0), (0, \pm 1)$, unless $a = b = c$, in which case there are 2 more primitive vectors, namely $u = (1, -1), (-1, 1)$.

Proof. Let $|y| \geq 2$. Then for any $x \in \mathbb{Z}$,

$$\begin{aligned} 4aq(x, y) &= (2ax + by)^2 - Dy^2 \geq 4|D| = 4(4ac - b^2) \\ \implies aq(x, y) &\geq 4ac - b^2. \end{aligned}$$

Now since $-a < b \leq a \leq c$, we have $b^2 \leq ac$ so $aq(x, y) \geq 4ac - ac = 3ac \implies q(x, y) \geq 3c > c$.

Next let $|y| = 1$, say $y = 1$ first. Then

$$4aq(x, 1) = (2ax + b)^2 - D$$

Notice that $|2ax + b| \geq |b|$ for all integers x . In particular $|2ax + b| = |b|$ if and only if $x = 0$ or $x = -1$ (when $a = b$). In all other cases, $|2ax + b| \geq 2a|x| - b \geq 2a - b \geq a$. Hence,

$$\begin{aligned} 4aq(x, 1) &\geq b^2 - D = 4ac \\ \implies q(x, 1) &\geq c. \end{aligned}$$

Similarly, for $y = -1$,

$$q(x, -1) \geq c.$$

Hence we have both the cases. □

38 Theorem

Let $q_1 = (a_1, b_1, c_1), q_2 = (a_2, b_2, c_2)$ be reduced binary quadratic forms of the same negative discriminant, with $q_1 \stackrel{+}{\sim} q_2$. Then $q_1 = q_2$, i.e., $a_1 = a_2, b_1 = b_2, c_1 = c_2$. In other words, the reduced form is unique.

Proof. Since $q_1 \stackrel{+}{\sim} q_2$, we can write $q_2 = q_1 \cdot A$ for some $A \in \text{SL}_2(\mathbb{Z})$. Since $q_1 \stackrel{+}{\sim} q_2$, they represent the same set of values and in particular, their smallest values are equal, i.e., $a_1 = a_2 = a$ (say).

Note that since q_1 is reduced, we have either $-a_1 < b_1 \leq a_1 < c_1$ or $0 \leq b_1 \leq a_1 = c_1$. Let us consider the two cases one by one.

- **Case I:** $a_1 < c_1$. In this case, $a_1 = q_1(u)$ has exactly two primitive solutions, namely $u = (\pm 1, 0)$. Hence $a = a_1 = a_2 = q_2(u)$ also has exactly two solutions, namely $u = (\pm 1, 0)$. This must imply $a_2 < c_2$, else if $a_2 = c_2$, there would be at least 4 solutions. Also, $a = q_1(\pm 1, 0) = q_2(\pm 1, 0) = (q_1 \cdot A)(\pm 1, 0) = q_1(\pm \alpha, \pm \gamma)$. By uniqueness, we get $\gamma = 0, \alpha = \pm 1$. $\alpha\delta - \gamma\beta = 1$ then implies $\delta = \pm 1$. Thus, we have

$$A = \begin{pmatrix} \pm 1 & \beta \\ 0 & \pm 1 \end{pmatrix}.$$

So, $(a_2, b_2, c_2) = q_1 \cdot A$. Expanding $q_1 \cdot A$, we get $b_2 = b_1 \pm 2a\beta$. Since q_1 and q_2 are reduced, we have $-a_1 < b_1 \leq a_1$ and $-a_2 < b_2 \leq a_2$, i.e., $-a \leq -b_1 < a$ and $-a < b_1 \pm 2a\beta \leq a$. Adding these two inequalities, we obtain $-2a < \pm 2a\beta < 2a$, forcing $\beta = 0$. Thus $A = \pm I$ and $q_2 = q_1(\pm I) = q_1$.

- **Case II:** $a_1 = c_1 = a$ (say). In this case, $q_1 = (a, b_1, a)$ so $b_1 \geq 0$ as q_1 is reduced. We now deal in subcases.

- A. $0 \leq b_1 < a$: In this subcase, $a = q_1(u)$ has exactly four primitive solutions $V = (\pm 1, 0), (0, \pm 1)$. Thus $a = q_2(w)$ will also have exactly four solutions. Hence $a_2 = c_2 > b_2 \geq 0$. Thus, $q_2 = (a, b_2, a)$. Since

$$\begin{aligned} b_2^2 - 4a^2 &= \text{disc} q_2 = \text{disc} q_1 = b_1^2 - 4a^2 \\ \implies b_1 &= \pm b_2. \end{aligned}$$

But both are non-negative, so we must have $b_1 = b_2$.

- B. $a_1 = b_1 = c_1$. Thus $q_1 = (a, a, a)$. Hence, $a = q_1(u)$ has six primitive solutions, namely $u = (\pm 1, 0), (0, \pm 1), (1, -1), (-1, 1)$. This must mean that $a = q_2(v)$ also has six primitive solutions. Thus $a_2 = b_2 = c_2 = a$. Thus $q_1 = q_2$.

□

39 Proposition

Let $q = (a, b, c)$ be a reduced binary quadratic form of discriminant $D = b^2 - 4ac < 0$. If $\gcd(x, y) = 1$ and $a \leq q(x, y) \leq c$. Then $q(x, y)$ must be a or c , i.e., $(x, y) = (\pm 1, 0)$ or $(0, \pm 1)$.

Proof. We know that $q(x, y) > c$ for $|y| \geq 2$. Hence $a \leq q(x, y) \leq c \implies |y| \leq 1$. The rest may be completed via a previous theorem. $\square$

40 Theorem

Let q be a binary quadratic form of discriminant $D < 0$. Then there is a reduced binary quadratic form (a, b, c) such that $q \sim^+ (a, b, c)$.

Existential proof. By the Lagrange Reduction Lemma (Theorem 25), there is a form (a, b, c) such that $q \sim^+ (a, b, c)$, and $|b| \leq |a| \leq |c|$. By our convention, we are in a positive definite setting and hence we have $a > 0, c > 0$. Thus $|b| \leq a \leq c$. Now, if $-a < b \leq a < c$, then we are done as (a, b, c) is then reduced. We look at the remaining two cases –

- **Case I:** $a = c$ but $b < 0$. In this case, $(a, b, c) = (a, b, a) \sim^+ (a, -b, a)$ via the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The form $(a, -b, a)$ is reduced in this case.
- **Case II:** $a < c$ but $b = -a$. In this case, $(a, b, c) = (a, -a, c) \sim^+ (a, -a + 2a, c - a + a) = (a, a, c)$ via the matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The form (a, a, c) is reduced.

Thus, in all cases, we end up with a reduced form properly equivalent to the given binary quadratic form. $\square$

Constructive proof. Let $q_1 = (a_0, b_0, c_0)$. If q_1 is reduced, we are simply done. If not, either $a_0 > c_0$ or $a_0 \leq c_0$ but $b_0 \notin (-a_0, a_0]$.

- **Case I:** $a_0 > c_0$. In this case, $q_1 = (a_0, b_0, c_0) \sim^+ (c_0, -b_0, a_0) = (a_1, b_1, c_1)$. Note that $|a_1| + |b_1| = |c_0| + |b_0| < |a_0| + |b_0|$.
- **Case II:** $a_0 \leq c_0$ but $b_0 \notin (-a_0, a_0]$. We deal with subcases.
 - A. $b_0 > a_0$. Then $(a_0, b_0, c_0) \sim^+ (a_0, b_0 - 2a_0, c_0 - b_0 + a_0) = (a_1, b_1, c_1)$. Note here that $|a_1| + |b_1| = |a_0| + |b_0 - 2a_0| < |a_0| + |b_0|$.
 - B. $b_0 < -a_0$. In this subcase, $(a_0, b_0, c_0) \sim^+ (a_0, b_0 + 2a_0, c_0 + b_0 + a_0) = (a_1, b_1, c_1)$. Note again here that $|a_1| + |b_1| = |a_0| + |b_0 + 2a_0| < |a_0| + |b_0|$.
 - C. $b_0 = -a_0$. In this subcase, $(a_0, b_0, c_0) = (a_0, -a_0, c_0) \sim^+ (a_0, a_0, c_0)$. In this subcase, the process terminates as we have already obtained a reduced binary quadratic form to which q is properly equivalent.

In general, continuing this way, we get a sequence $q_1 = (a_0, b_0, c_0) \stackrel{+}{\sim} (a_1, b_1, c_1) \stackrel{+}{\sim} \dots \stackrel{+}{\sim} (a_j, b_j, c_j)$ with $|a_0| + |b_0| > |a_1| + |b_1| > \dots > |a_j| + |b_j| > 0$. This process must stop at a finite number of steps, say at (a_j, b_j, c_j) . This means that (a_j, b_j, c_j) is either reduced or of the form $(a_j, -a_j, c_j)$, which is properly equivalent to (a_j, a_j, c_j) , a reduced form. $\square$

§6 Some further terminology

We sometimes encounter the following terminology in the literature.

A binary quadratic form $q = (a, b, c)$ is said to be *positive semidefinite* if

$$q(x, y) \geq 0 \quad \text{for all } x, y \in \mathbb{Z}.$$

It is called *positive definite* if

$$q(x, y) \geq 0 \quad \text{for all } x, y \in \mathbb{Z}, \quad q(x, y) = 0 \Rightarrow x = y = 0.$$

Similarly one defines *negative semidefinite* and *negative definite* quadratic forms. A form is called *indefinite* if it is neither positive nor negative semidefinite. These properties can also be characterized in terms of the discriminant of the quadratic form.

§7 Exercises

Exercises for Chapter 14

1. Write $a_n + b_n\sqrt{2} = (1 + \sqrt{2})^n$. Prove that

$$a_n^2 - 2b_n^2 = (-1)^n$$

for all $n \geq 1$. Conclude that the binary quadratic form $X^2 - 2Y^2$ represents the value 1 infinitely many times. This is in stark contrast with the positive definite case, where a given number is represented at most a finite number of times.

2. Prove the following identity of Brahmagupta: For any integers a, b, c, d, D , we have

$$(a^2 + Db^2)(c^2 + Dd^2) = (ac - Dbd)^2 + D(ad + bc)^2.$$

3. Determine whether the following pairs of binary quadratic forms are properly equivalent or not. If they are properly equivalent, give a matrix which takes one to the other:

- $(1, 2, 3)$ and $(1, 0, 2)$
- $(1, 1, 3)$ and $(1, -1, 3)$.
- $(1, 0, 3)$ and $(3, 0, 1)$.
- $(2, 3, 4)$ and $(2, -3, 4)$.
- $(3, 2, 1)$ and $(1, 2, 3)$.
- $(1, 3, 10)$ and $(2, 3, 5)$.

4. Prove that $h(D) = 1$ for each of the following discriminants (**Gauss' Theorem**):

$$D \in \{-3, -4, -7, -8, -11, -12, -16, -19, -27, -28, -43, -67, -163\}.$$

5. Compute the class number $h(D)$ for each of the following values of D :

- | | |
|-------------|-------------|
| (a) -15 . | (d) -59 . |
| (b) 13 . | (e) -100 |
| (c) 21 . | (f) -47 . |

6. Describe the primes p for which there are integers x and y such that $x^2 + xy + 7y^2 = p$. Give examples of such primes $p > 100$ with the corresponding values of x and y .

7. Find the reduced forms of the following binary quadratic forms:

- | | |
|------------------|-----------------|
| (a) $(2, 6, 9)$ | (d) $(1, 3, 3)$ |
| (b) $(3, -1, 3)$ | (e) $(2, 6, 7)$ |
| (c) $(5, 5, 4)$ | (f) $(1, 0, 1)$ |

8. Let p be a prime and $f(X, Y)$ be a primitive binary quadratic form. Prove that p cannot divide all of $f(0, 1)$, $f(1, 0)$ and $f(1, 1)$. Deduce that if S is a finite set of primes, there are coprime integers x and y such that $p \nmid f(x, y)$ for any $p \in S$.

9. Show that $X^2 + 5Y^2$ and $2X^2 + 2XY + 3Y^2$ are the only reduced binary quadratic forms of discriminant -20 .

- i. Prove that the first of these forms does not represent 2, but that the second one does. Deduce that the class number $h(-20) = 2$.
- ii. Show that an odd prime p is represented by at least one of these forms if and only if $p = 5$ or $p \equiv 1, 3, 7, \text{ or } 9 \pmod{20}$.

10. Find all reduced binary quadratic forms (a, b, c) such that $a > 0$ and $b^2 - 4ac = 69$.
11. Find all reduced binary quadratic forms (a, b, c) such that $a > 0$ and $b^2 - 4ac = 77$.
12. Find all reduced binary quadratic forms (a, b, c) such that $a > 0$ and $b^2 - 4ac = 193$.
13. Compute $h(-47)$.